Similar questions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A mass m is attached to a light spring of stiffness κ, where κ is the force needed per unit extension (units $\mathrm{N\,m^{-1}}$). The mass oscillates, and the time for one complete oscillation is denoted by τ. A student suggests the model $\tau = k m^{\alpha} \kappa^{\beta}$, where k is a dimensionless constant.

(a) Find the dimensions of κ. **[1]**

(b) Use dimensional analysis to find α and β. **[4]**

(c) Find the effect on τ of multiplying m by 4 and κ by 9 simultaneously. **[2]**

(d) The student finds experimentally that the amplitude of the oscillation has no effect on τ. Explain how this is consistent with the model. **[1]**

2 The force F exerted by wind on a sail is modelled by $F = k\rho^{\alpha} v^{\beta} A^{\gamma}$, where ρ is the density of air, v is the wind speed relative to the sail, A is the area of the sail and k is a dimensionless constant.

(a) Find α, β and γ. **[5]**

(b) State the effect on F of doubling the wind speed and halving the area. **[2]**

It is suggested that the viscosity μ of the air should also be included, with dimensions $\mathrm{ML^{-1}T^{-1}}$, so that $F = k\rho^{\alpha} v^{\beta} A^{\gamma} \mu^{\delta}$.

(c) Explain why dimensional analysis alone cannot determine all four indices. **[1]**

(d) Given that $\beta = 1$, find α, γ and δ. **[4]**

3 A student models the speed v of a wave travelling along a stretched string by $v = kT^{\alpha} \mu^{\beta} L^{\gamma}$, where T is the tension in the string, μ is the mass per unit length of the string, L is the length of the string and k is a dimensionless constant.

(a) Find the dimensions of μ. **[1]**

(b) Determine α, β and γ. **[5]**

(c) Comment on what the value of γ says about the effect of the length of the string. **[1]**

(d) Find the factor by which the tension must be multiplied to double the wave speed. **[2]**

(e) Another student proposes $v = k\sqrt{TL/\mu}$. Show that this formula is dimensionally inconsistent. **[2]**

4 A satellite of mass m moves in a circular orbit of radius r about a planet of mass M. The universal gravitational constant G has dimensions $\mathrm{M^{-1}L^{3}T^{-2}}$. The period τ of the orbit is modelled by $\tau = kG^{\alpha} M^{\beta} r^{\gamma}$, where k is a dimensionless constant (the mass m of the satellite is not included).

(a) Use dimensional analysis to find α, β and γ. **[5]**

(b) Find the effect on the period of multiplying the radius of the orbit by 4. **[2]**

(c) Explain why, if m were included as well with an unknown index δ, dimensional analysis alone could not determine β and δ. **[2]**

(d) A student suggests that the escape speed from the planet is $v = k\sqrt{GMr}$. Show by dimensional analysis that this cannot be correct, and write down a correct form. **[3]**

Question 1
(a) $[\kappa] = \mathrm{MLT}^{-2}/\mathrm{L} = \mathrm{MT}^{-2}$
(b) $\mathrm{T} = \mathrm{M}^{\alpha}(\mathrm{MT}^{-2})^{\beta}$. Powers of M: $\alpha+\beta=0$. Powers of T: $1=-2\beta$. So $\beta=-\frac{1}{2}$ and $\alpha=\frac{1}{2}$, i.e. $\tau = k\sqrt{m/\kappa}$
(c) τ is multiplied by $\sqrt{4/9}=\frac{2}{3}$
(d) Amplitude (a length) does not appear in the model, so the model predicts that it has no effect

Question 2
(a) $\mathrm{MLT}^{-2} = (\mathrm{ML}^{-3})^{\alpha}(\mathrm{LT}^{-1})^{\beta}(\mathrm{L}^2)^{\gamma}$. M: $\alpha=1$. T: $\beta=2$. L: $1=-3\alpha+\beta+2\gamma=-1+2\gamma$, so $\gamma=1$. $F=k\rho v^2 A$
(b) F is multiplied by $2^2\times\frac{1}{2}=2$
(c) There are only three independent dimensions (M, L, T) giving three equations for four unknowns
(d) T: $-2=-\beta-\delta$, so $\delta=1$. M: $1=\alpha+\delta$, so $\alpha=0$. L: $1=-3\alpha+\beta+2\gamma-\delta=1+2\gamma-1$, so $\gamma=\frac{1}{2}$. So $F=kv\mu\sqrt{A}$

Question 3
(a) $[\mu] = \mathrm{ML}^{-1}$
(b) $\mathrm{LT}^{-1} = (\mathrm{MLT}^{-2})^{\alpha}(\mathrm{ML}^{-1})^{\beta}\mathrm{L}^{\gamma}$. T: $-1=-2\alpha$, $\alpha=\frac{1}{2}$. M: $0=\alpha+\beta$, $\beta=-\frac{1}{2}$. L: $1=\alpha-\beta+\gamma=1+\gamma$, so $\gamma=0$
(c) $\gamma=0$: the length of the string has no effect on the wave speed
(d) $v\propto\sqrt{T}$, so T must be multiplied by 4
(e) $[TL/\mu] = \mathrm{MLT}^{-2}\cdot\mathrm{L}\cdot\mathrm{M}^{-1}\mathrm{L} = \mathrm{L}^3\mathrm{T}^{-2}$, so the root has dimensions $\mathrm{L}^{3/2}\mathrm{T}^{-1}\neq\mathrm{LT}^{-1}$

Question 4
(a) $\mathrm{T} = (\mathrm{M}^{-1}\mathrm{L}^3\mathrm{T}^{-2})^{\alpha}\mathrm{M}^{\beta}\mathrm{L}^{\gamma}$. T: $1=-2\alpha$, $\alpha=-\frac{1}{2}$. M: $0=-\alpha+\beta$, $\beta=-\frac{1}{2}$. L: $0=3\alpha+\gamma$, $\gamma=\frac{3}{2}$. So $\tau = kr^{3/2}/\sqrt{GM}$
(b) τ is multiplied by $4^{3/2}=8$
(c) The M equation would become $0=-\alpha+\beta+\delta$ with $\alpha=-\frac{1}{2}$ fixed, so only $\beta+\delta=-\frac{1}{2}$ is determined: two unknowns, one equation. (Physically $\delta=0$.)
(d) $[GMr] = \mathrm{L}^3\mathrm{T}^{-2}\cdot\mathrm{L} = \mathrm{L}^4\mathrm{T}^{-2}$, so $\sqrt{GMr}$ has dimensions $\mathrm{L}^2\mathrm{T}^{-1}\neq\mathrm{LT}^{-1}$. A correct form is $v=k\sqrt{GM/r}$